

Convex sets and functions

1 Introduction

“Convexity” in various forms is important in economics. We begin with the basic building block, a convex set. All subsequent definitions, including those that relate to functions, can be traced back to this *one* building block. At *no point* in these notes will we impose differentiability or examine second order derivatives. Thus, it is emphasized that convexity (or concavity) of functions is a very fundamental concept and one that does not require more assumptions such as differentiability. In this sense, differentiability can be a distraction and one that is not needed to fundamentally understand convexity.

More generally, these notes illustrate an important point that applies to all theoretical work: The ideal is to work with as few assumptions as possible. When we impose extra assumptions in practice, such as differentiability, it is often for convenience (for example, differentiability allows the use of calculus) rather than for good scientific reasons. Moreover, theorists enjoy finding or exposing patterns and making connections between different concepts.

Some parts of these notes are covered in Appendices M.F, M.G, and M.C of MWG. There are entire textbooks devoted just to convexity (but you will not need such detailed knowledge). A classic but very advanced reference is Rockafellar (1970); a more approachable textbook is Boyd and Vandenberghe (2004). See Vohra (2005) for a compact textbook covering convexity and a variety of other subjects.

It is assumed that you know what open, closed, and compact sets are, and what the interior of a set, and its closure, are. A refresher is in the Appendix at the end of these notes. The Appendix also contains an (optional) introduction to hyperplanes, which will prove useful later on. This part also supplements (or explains in more detail) the technical Section 3.F in MWG on duality.

2 Convex sets

We consider sets in $\mathbb{R}^n$. Let $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ and $\mathbf{x}' = (x'_1, x'_2, \dots, x'_n) \in \mathbb{R}^n$ be two points in $\mathbb{R}^n$. For any $t \in [0, 1]$, let $\mathbf{x}^t = (x_1^t, x_2^t, \dots, x_n^t)$ be the *convex combination* (or weighted average) of the two points, such that

$$x_i^t = tx_i + (1 - t)x'_i \text{ for all } i = 1, 2, \dots, n.$$

Convex set: Let $S \subseteq \mathbb{R}^n$ be some set in $\mathbb{R}^n$. The set S is a convex set if for all $\mathbf{x}, \mathbf{x}' \in S$ and all $t \in [0, 1]$, it holds that $\mathbf{x}^t \in S$. In words, the average of any two points in the set is also in the set.

Strictly convex set: Let $S \subseteq \mathbb{R}^n$ be some set in $\mathbb{R}^n$. The set S is a strictly convex set if for all $\mathbf{x}, \mathbf{x}' \in S$ with $\mathbf{x} \neq \mathbf{x}'$ and all $t \in (0, 1)$, it holds that $\mathbf{x}^t \in \text{int}S$. In words, $\mathbf{x}^t$ is in the interior of S , meaning that an open ball around $\mathbf{x}^t$ is also in S .

A basic fact is that *the intersection of convex sets is convex*: Assume $\mathbf{x}, \mathbf{x}' \in S \cap T$, meaning that $\mathbf{x}$ and $\mathbf{x}'$ belong to both S and T . If S and T are convex sets, then $\mathbf{x}^t$ belongs to both S and T , or $\mathbf{x}^t \in S \cap T$.

Convex sets have the property that a convex combination of *any number of points* in S is also in S . That is, $\sum_{i=1}^m t_i \mathbf{x}_i \in S$ whenever $\mathbf{x}_i \in S$ and $t_i \geq 0$ for all $i = 1, \dots, m$, and $\sum_{i=1}^m t_i = 1$. It turns out that there are specific points in S that are especially useful to consider.

Extreme points: Let $S \subseteq \mathbb{R}^n$ be some convex set in $\mathbb{R}^n$. Then, $\mathbf{x} \in S$ is an extreme point of S if it cannot be expressed as a convex combination of two distinct points in S , i.e. it *cannot* be obtained as $\mathbf{x} = t\mathbf{x}' + (1-t)\mathbf{x}''$ for any $\mathbf{x}', \mathbf{x}'' \in S$, with $\mathbf{x}' \neq \mathbf{x}''$, and $t \in (0, 1)$.

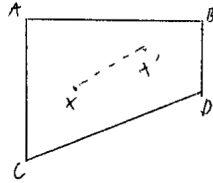
Theorem: Let $S \subseteq \mathbb{R}^n$ be some convex set in $\mathbb{R}^n$. Assume S is compact (i.e. closed and bounded). Then, every $\mathbf{x} \in S$ is a convex combination of at most $n + 1$ extreme points in S .

Of course, not all sets are convex sets. Sometimes, we want to describe the smallest convex set that contains a non-convex set. This idea leads to the following definition.

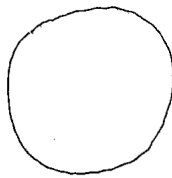
The convex hull: Let $S \subseteq \mathbb{R}^n$ be some set in $\mathbb{R}^n$ (possibly not convex). The convex hull of S , $\text{co}S$ (or sometimes $\text{conv}S$), is the intersection of all convex sets that contain S . This is a convex set because it is the intersection of convex sets, and it is the unique smallest set that contains S .

It can be shown that $\text{co}S$ can equivalently be described as the set of all convex combinations of elements of S . In fact, it can be obtained by taking convex combinations of at most $n+1$ elements at a time. This famous result is known as Carathéodory's Theorem. Similarly, and in line with the above theorem, if S is a compact and convex set, then it is the convex hull of its extreme points (the Krein–Milman Theorem). Thus, if we know the extreme points, then we have enough information to recover the entire set.

Some graphical examples are on the next page. The figures in these notes follow the convention that open circles, $\circ$, and dashed lines signify non-inclusion, while filled-in circles, $\bullet$, and solid lines signify inclusion. This enables us to graphically depict the difference between open and closed sets.



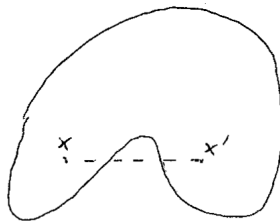
A COMPACT AND CONVEX SET THAT
IS NOT STRICTLY CONVEX AND
WHICH HAS 4 EXTREME POINTS (A, B, C, D)



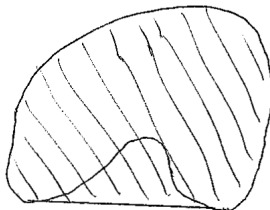
A COMPACT AND STRICTLY CONVEX
SET WITH INFINITELY MANY
EXTREME POINTS (= THE BOUNDARY)



AN OPEN, BOUNDED, AND STRICTLY
CONVEX SET WITH NO EXTREME
POINTS



A NON-CONVEX SET



THE CONVEX HULL OF A
NON-CONVEX SET

3 Convex functions

3.1 Some more sets

Sets also play an important role in our definition of functions. Thus, we start by identifying a number of important sets.

Consider some *domain* $X \subseteq \mathbb{R}^n$. This is the *set* of possible inputs of the *function* that we consider. A function $f : X \rightarrow \mathbb{R}$ assigns a *single output value* in $\mathbb{R}$ to each $\mathbf{x} \in X$, written $f(\mathbf{x})$. It is often the case in economics that the domain is a “product set”, in which case it can be written as $X = X_1 \times X_2 \times \dots \times X_n$ or $X = \times_{i=1}^n X_i$, to denote that each x_i must belong to the corresponding set X_i . If each X_i is convex, then X is convex. If each X_i is an interval, then X is a rectangle ($n = 2$), box ($n = 3$) or more generally a hyper-rectangle. Product sets are special because the value of x_i does not influence which values of x_j are feasible, $i \neq j$. This is not the case if the domain is e.g. a circle or a ball.

The *image* (or sometimes range) of the function is the *set* of all possible output values, $Y = \{y \in \mathbb{R} | y = f(\mathbf{x}) \text{ for some } \mathbf{x} \in X\}$. The *graph* of the function is the combination of inputs and outputs that are possible, i.e. it is the *set*

$$G(f) = \{(\mathbf{x}, y) \in \mathbb{R}^{n+1} | \mathbf{x} \in X \text{ and } y = f(\mathbf{x})\},$$

which is a subset of $X \times Y$, which is itself a subset of $\mathbb{R}^{n+1}$. The graph is what we usually draw when we want to visualize a function (for $n = 1$ or $n = 2$ we draw 2D or 3D pictures, respectively).

The next concepts are less commonly used, but they are also important to understand the role of appropriately defined sets in determining various properties of functions. The *epigraph* of a function is the set of points in $\mathbb{R}^{n+1}$ that *lies on or above the graph* (epi means above). Formally, it is the *set*

$$\text{epi}(f) = \{(\mathbf{x}, y) \in \mathbb{R}^{n+1} | \mathbf{x} \in X \text{ and } y \geq f(\mathbf{x})\}.$$

Similarly, the *hypograph* is the *set* of points on or below the graph, i.e. it is the set

$$\text{hypo}(f) = \{(\mathbf{x}, y) \in \mathbb{R}^{n+1} | \mathbf{x} \in X \text{ and } y \leq f(\mathbf{x})\}.$$

In addition to using the graph to visualize a function, we often use indifference curves, which is a depiction of the level set. A level set (at level k) is a set of the form $\{\mathbf{x} \in X | f(\mathbf{x}) = k\}$, describing the set of inputs that give rise to a specific output value (k). The *superior set* or upper contour set is the set of inputs that give weakly higher output values, or $\{\mathbf{x} \in X | f(\mathbf{x}) \geq k\}$. Similarly, the *inferior set* or lower contour set is $\{\mathbf{x} \in X | f(\mathbf{x}) \leq k\}$. These are sets in $\mathbb{R}^n$, so they have lower dimensionality than the graph, epigraph, and hypograph.

3.2 Functions and relationship to sets

We are now ready to define convex functions and other related functions. To do so, assume from now on that X is a convex set.

Convex function: Let $X \subseteq \mathbb{R}^n$ be some convex set in $\mathbb{R}^n$ and let $f : X \rightarrow \mathbb{R}$. The function f is convex if for all $\mathbf{x}, \mathbf{x}' \in X$ and all $t \in [0, 1]$, it holds that

$$f(\mathbf{x}^t) \leq tf(\mathbf{x}) + (1 - t)f(\mathbf{x}').$$

In words, the function-value of an average point is less than the average of the function-values of the original points. Equivalently, *the epigraph is a convex set*.

Quasiconvex function: Let $X \subseteq \mathbb{R}^n$ be some convex set in $\mathbb{R}^n$ and let $f : X \rightarrow \mathbb{R}$. The function f is quasiconvex if for all $\mathbf{x}, \mathbf{x}' \in X$ and all $t \in [0, 1]$, it holds that

$$f(\mathbf{x}^t) \leq \max\{f(\mathbf{x}), f(\mathbf{x}')\}.$$

In words, the function-value of an average point is less than the highest function-value of the original points. Equivalently, *the inferior set is a convex set*.

The function f is called *strictly* convex or strictly quasiconvex, respectively, if the weak inequality is replaced with a strict inequality when $\mathbf{x} \neq \mathbf{x}'$ and $t \in (0, 1)$.

There is no such thing as a concave set. When we talk of concave and quasiconcave functions, we mean that $-f$ is convex or quasiconvex, respectively. The following definitions are equivalent.

Concave function: Let $X \subseteq \mathbb{R}^n$ be some convex set in $\mathbb{R}^n$ and let $f : X \rightarrow \mathbb{R}$. The function f is concave if for all $\mathbf{x}, \mathbf{x}' \in X$ and all $t \in [0, 1]$, it holds that

$$f(\mathbf{x}^t) \geq tf(\mathbf{x}) + (1 - t)f(\mathbf{x}').$$

In words, the function-value of an average point is less than the average of the function-values of the original points. Equivalently, *the hypograph is a convex set*.

Quasiconcave function: Let $X \subseteq \mathbb{R}^n$ be some convex set in $\mathbb{R}^n$ and let $f : X \rightarrow \mathbb{R}$. The function f is quasiconcave if for all $\mathbf{x}, \mathbf{x}' \in X$ and all $t \in [0, 1]$, it holds that

$$f(\mathbf{x}^t) \geq \min\{f(\mathbf{x}), f(\mathbf{x}')\},$$

or equivalently if *the superior set is a convex set*.

Note that since

$$\max\{f(\mathbf{x}), f(\mathbf{x}')\} \geq tf(\mathbf{x}) + (1 - t)f(\mathbf{x}') \geq \min\{f(\mathbf{x}), f(\mathbf{x}')\},$$

any convex function is quasiconvex and any concave function is quasiconcave. It is also the case that the sum of convex (concave) functions is convex (concave), but this does not extend to quasiconvex or quasiconcave functions.

Finally, note that if you want to determine if a function is quasiconcave, it is not sufficient to sketch the indifference curve and determine if it is convex. You need to locate the superior set, but the location of the superior set depends on whether $f(\mathbf{x})$ is increasing or decreasing in the variables.

Exercise: Let $X = \mathbb{R}_+^2$ and consider the functions $f(\mathbf{x}) = x_1x_2$ and $g(\mathbf{x}) = -x_1x_2$. Sketch indifference curves for the two functions. Is $f(\mathbf{x})$ quasiconcave? What about $g(\mathbf{x})$?

Exercise: Let $X = \mathbb{R}$ and draw the graph of $f(x) = |x|$ (the absolute value of x). Verify visually that the epigraph is convex. How do you identify the inferior set in your picture? Is it convex?

3.3 Implication/applications

A small selection of implications or application follows.

Implication 1: Let $h(\mathbf{x})$ and $g(\mathbf{x})$ be convex functions and consider the function $f(\mathbf{x}) = \max\{h(\mathbf{x}), g(\mathbf{x})\}$. Note that

$$\begin{aligned} f(\mathbf{x}^t) &= \max\{h(\mathbf{x}^t), g(\mathbf{x}^t)\} \\ &\leq \max\{th(\mathbf{x}) + (1-t)h(\mathbf{x}'), tg(\mathbf{x}) + (1-t)g(\mathbf{x}')\} \text{ since } h \text{ and } g \text{ are convex} \\ &\leq t \max\{h(\mathbf{x}), g(\mathbf{x})\} + (1-t) \max\{h(\mathbf{x}'), g(\mathbf{x}')\} \\ &= tf(\mathbf{x}) + (1-t)f(\mathbf{x}'), \end{aligned}$$

which implies that f is a convex function. An alternative proof relies on examining the properties of the epigraph,

$$\begin{aligned} \text{epi}(f) &= \{(\mathbf{x}, y) \in \mathbb{R}^{n+1} | \mathbf{x} \in X \text{ and } y \geq \max\{h(\mathbf{x}), g(\mathbf{x})\}\} \\ &= \{(\mathbf{x}, y) \in \mathbb{R}^{n+1} | \mathbf{x} \in X \text{ and } y \geq h(\mathbf{x}) \text{ and } y \geq g(\mathbf{x})\} \\ &= \{(\mathbf{x}, y) \in \mathbb{R}^{n+1} | \mathbf{x} \in X \text{ and } y \geq h(\mathbf{x})\} \cap \{(\mathbf{x}, y) \in \mathbb{R}^{n+1} | \mathbf{x} \in X \text{ and } y \geq g(\mathbf{x})\} \\ &= \text{epi}(h) \cap \text{epi}(g) \end{aligned}$$

Since $\text{epi}(h)$ and $\text{epi}(g)$ are convex sets, $\text{epi}(f) = \text{epi}(h) \cap \text{epi}(g)$ is a convex set as well and it follows that f is a convex function. For similar reasons, $f(\mathbf{x}) = \min\{h(\mathbf{x}), g(\mathbf{x})\}$ is a concave function if $h(\mathbf{x})$ and $g(\mathbf{x})$ are concave.

Implication 2: Let h and g be two different functions. The domain of g is a convex subset of $\mathbb{R}^n$. The domain of h is a subset of $\mathbb{R}$ that contains the range of g . We are

interested in the composite function $f(\mathbf{x}) = h(g(\mathbf{x}))$. Assume that g and h are both concave, and that h is non-decreasing. Then, f is concave as well. To see this, note that

$$\begin{aligned} f(\mathbf{x}^t) &= h(g(\mathbf{x}^t)) \\ &\geq h(tg(\mathbf{x}) + (1-t)g(\mathbf{x}')) \\ &\geq th(g(\mathbf{x})) + (1-t)h(g(\mathbf{x}')) \\ &= tf(\mathbf{x}) + (1-t)f(\mathbf{x}'). \end{aligned}$$

The first inequality follows from the fact that g is concave and h is non-decreasing. The second inequality follows because h is concave and $tg(\mathbf{x}) + (1-t)g(\mathbf{x}')$ is the convex combination between $g(\mathbf{x})$ and $g(\mathbf{x}')$.

The intuition is that we started with a concave function, g , and then subjected it to a concave transformation, h , in order to make it “even more” concave. However, this intuition is valid only if the transformation is also non-decreasing. A non-decreasing and concave transformation that is commonly used in economics is $h(g) = \ln g$.

More generally, note that even if $g(\mathbf{x})$ is not concave, $h(g(\mathbf{x}))$ could be, provided h is “concave enough” to override or dominate the curvature of $g(\mathbf{x})$. In the special case where $\ln g(\mathbf{x})$ is concave, we commonly refer to $g(\mathbf{x})$ as *log-concave*. The product of log-concave functions is itself a log-concave function (can you see why?).

Remark: Concavity is a *cardinal property* (it is sensitive to how we measure things). That is, if $h(\cdot)$ is an increasing (and therefore order-preserving) transformation, then $h(g(\mathbf{x}))$ may be concave even if $g(\mathbf{x})$ is not, and vice versa. In contrast, quasiconcavity is an *ordinal property* (not sensitive to how we measure things) – for any increasing transformation, $h(g(\mathbf{x}))$ is quasiconcave if and only if $g(\mathbf{x})$ is.

A useful implication is that if we can find *some* increasing transformation that makes $h(g(\mathbf{x}))$ concave, then $h(g(\mathbf{x}))$ is quasiconcave, which means that $g(\mathbf{x})$ is quasiconcave. So, to check if $g(\mathbf{x})$ is quasiconcave, it is sometimes easier to see if you can find a transformation that makes $h(g(\mathbf{x}))$ concave. For example, $g(\mathbf{x}) = e^{-(x_1+x_2)^2}$ is not concave, but if $h(g) = \ln g$, then $h(g(\mathbf{x})) = -(x_1 + x_2)^2$ is concave. Hence, $g(\mathbf{x})$ is quasiconcave.

Implication 3: The famous and highly useful *Jensen’s inequality* says that the function f is convex if and only if

$$f(\sum_{i=1}^m t_i \mathbf{x}_i) \leq \sum_{i=1}^m t_i f(\mathbf{x}_i) \text{ for all } t_1, t_2, \dots, t_m \geq 0 \text{ with } \sum_{i=1}^m t_i = 1$$

(the inequality is reversed if the function is concave). To understand Jensen’s inequality, recall from our discussion of convex sets that a set S is convex if and only if a convex combination of any number of points in S is also in S . That is, $\sum_{i=1}^m t_i \mathbf{z}_i \in S$ whenever $\mathbf{z}_i \in S$ for all $t_1, t_2, \dots, t_m \geq 0$ with $\sum_{i=1}^m t_i = 1$. Now take S to be the epigraph of f , or $S = \text{epi}(f)$. Let $\mathbf{z}_i = (\mathbf{x}_i, y_i)$ with $\mathbf{x}_i \in X$ and $y_i \geq f(\mathbf{x}_i)$. Note that $\mathbf{z}_i \in S = \text{epi}(f)$ by definition.

Assume first that f is convex, or in other words that $S = \text{epi}(f)$ is a convex set. Thus, convex combinations of $(\mathbf{x}_i, y_i)$ is in the epigraph, or $\sum_{i=1}^m t_i(\mathbf{x}_i, y_i) \in \text{epi}(f)$. That is,

$$(\sum_{i=1}^m t_i \mathbf{x}_i, \sum_{i=1}^m t_i y_i) \in \text{epi}(f),$$

which means that $\sum_{i=1}^m t_i y_i \geq f(\sum_{i=1}^m t_i \mathbf{x}_i)$. Jensen's inequality then follows by letting $y_i = f(\mathbf{x}_i)$. The reverse direction starts by assuming $f(\sum_{i=1}^m t_i \mathbf{x}_i) \leq \sum_{i=1}^m t_i f(\mathbf{x}_i)$, which in turn implies that $f(\sum_{i=1}^m t_i \mathbf{x}_i) \leq \sum_{i=1}^m t_i f(\mathbf{x}_i) \leq \sum_{i=1}^m t_i y_i$ whenever $y_i \geq f(\mathbf{x}_i)$ for all i . Thus, the convex combinations of $(\mathbf{x}_i, y_i)$ is in $\text{epi}(f)$, which then means that f is convex.

Remark: To reiterate, we have not assumed that the functions we study are differentiable. The point is that the basic definition of convexity or concavity does not require differentiability. Indeed, the examples above demonstrate that we can obtain many useful results though the basic definitions. In fact, in Implication 1, the function $f(\mathbf{x}) = \max\{h(\mathbf{x}), g(\mathbf{x})\}$ is typically not differentiable even when both $h(\mathbf{x})$ and $g(\mathbf{x})$ are differentiable.

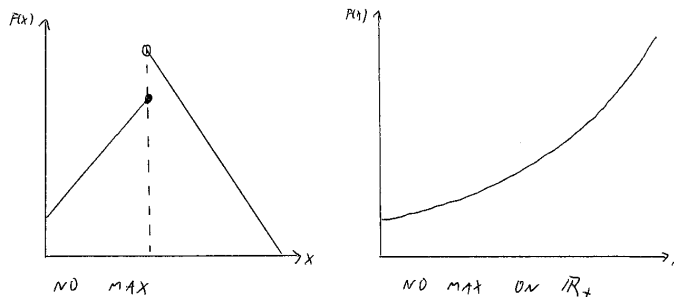
4 Optimization

4.1 Existence and uniqueness of solution(s)

In economics, we habitually encounter optimization problems such as $\min_{\mathbf{x} \in C} f(\mathbf{x})$ and $\max_{\mathbf{x} \in C} f(\mathbf{x})$, where $C \subseteq X$ is a “feasible set” or “constraint set”. A very basic question thus arises: When does a solution to an optimization problem exist? The first point that we should make is that we need very particular assumptions for existence to be guaranteed.

The Extreme Value Theorem: Consider the function $f : X \rightarrow \mathbb{R}$. If f is *continuous* and $C \subseteq X$ is non-empty and *compact* (closed and bounded), then there exist solutions to $\min_{\mathbf{x} \in C} f(\mathbf{x})$ and $\max_{\mathbf{x} \in C} f(\mathbf{x})$.

The Extreme Value Theorem is also known as the Weierstrass Maximum Theorem or the Maximum Value Theorem. The following figures illustrate why continuity and compactness are necessary for functions with domain in $\mathbb{R}$.



Example: Even if we restrict attention to functions that are nicely behaved in the sense that they are convex/concave or quasiconvex/quasiconcave, continuity is not guaranteed. For instance, the function

$$f(x) = \begin{cases} \sqrt{x} & \text{if } x \in [0, 1) \\ 0 & \text{if } x = 1 \end{cases},$$

with compact domain $X = [0, 1]$ is concave (check the definition), but it has no maximum. This example illustrates that concavity or convexity is not enough to guarantee continuity. However, it can be shown that any concave or convex function is continuous in the interior of its domain, $\text{int}X$. Indeed, it is differentiable almost everywhere on $\text{int}X$. For convex functions of one variable, an implication is that they are non-differentiable at at most countably many point. In contrast, quasiconcave and quasiconvex functions need not even be continuous in the interior.

Remark: This is a good place to remind yourself of the difference between the infimum and minimum on the one hand and the supremum and maximum on the other. The infimum (greatest lower bound) and supremum (least upper bound) of a function are always well-defined (but they may be $-\infty$ or ∞). In the previous example, the supremum is 1, the maximum does not exist, the infimum is 0, and the minimum exists and is 0 as well.

Moving on, let us assume one or more solutions exist, and ask what we can say about the structure of the set of solutions. In light of the previous theorem, it is perhaps not surprising that we need assumptions both on C and on f . For C , we assume convexity. Concerning f , it happily turns out that quasiconvexity/quasiconcavity is sufficient to be very precise. To read the theorem, recall that $\arg \min_{\mathbf{x} \in C} f(\mathbf{x})$ denotes the (set of) value(s) of $\mathbf{x} \in C$ that solve the problem $\min_{\mathbf{x} \in C} f(\mathbf{x})$.

Theorem: Consider the function $f : X \rightarrow \mathbb{R}$. Assume that $C \subseteq X$ is non-empty and convex. Assume that a solution to the optimization problem exists. Then:

1. If f is quasiconvex then $\arg \min_{\mathbf{x} \in C} f(\mathbf{x})$ is a convex set. If f is strictly quasiconvex then $\arg \min_{\mathbf{x} \in C} f(\mathbf{x})$ is unique.
2. If f is quasiconcave then $\arg \max_{\mathbf{x} \in C} f(\mathbf{x})$ is a convex set. If f is strictly quasiconcave then $\arg \max_{\mathbf{x} \in C} f(\mathbf{x})$ is unique.

The theorem shows that solutions, if there are more than one, are “bunched together”. Let us prove the second part, where f is quasiconcave (the first part is similar). Assume that $\mathbf{x}'$ and $\mathbf{x}''$ are both solutions to $\max_{\mathbf{x} \in C} f(\mathbf{x})$. Let v denote the maximum value, $\max_{\mathbf{x} \in X} f(\mathbf{x})$. Thus, $f(\mathbf{x}') = f(\mathbf{x}'') = v$. Since the set C is convex,

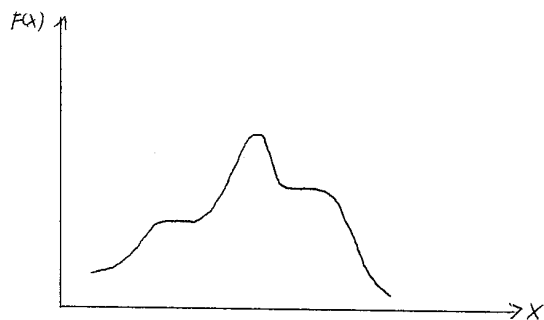
the convex combination of $\mathbf{x}'$ and $\mathbf{x}''$ also belong to C . Given f is quasiconcave, it holds that

$$\begin{aligned} f(\mathbf{x}^t) &\geq \min\{f(\mathbf{x}'), f(\mathbf{x}'')\} \\ &= \min\{v, v\} = v. \end{aligned}$$

However, it is impossible that $f(\mathbf{x}^t) > v$, because this would violate the assumption that $\mathbf{x}'$ and $\mathbf{x}''$ are solutions. Thus, $f(\mathbf{x}^t) = v$, which means that $\mathbf{x}^t$ is a solution too. In conclusion, if $\mathbf{x}'$ and $\mathbf{x}''$ solve the problem, then so does their convex combination.

The proof of uniqueness under strict quasiconcavity is similar, but let us spell it out. The purpose is to demonstrate a very important proof technique, namely *proof by contradiction*. Thus, we assume, in contradiction of what we want to prove, that there is more than one solution. Let $\mathbf{x}'$ and $\mathbf{x}''$ denote two distinct solution. Then, under strict quasiconcavity, $f(\mathbf{x}^t) > \min\{f(\mathbf{x}'), f(\mathbf{x}'')\}$. However, the strict inequality contradicts that $\mathbf{x}'$ and $\mathbf{x}''$ are solutions. Hence, there is at most one solution.

The intuition behind the theorem is simple, and important. If you visualize a concave function, you see a function with at most one “peak”. The peak is possibly at a plateau, in which case there are several solutions to $\max_{\mathbf{x} \in C} f(\mathbf{x})$. Now, the important point is that quasiconcave functions have the same feature. In particular, it is easy to see that if $X \subseteq \mathbb{R}$, then a quasiconcave function is *single-peaked*, as the example in the next figure illustrates. However, note that the function in the figure has stationary points that are not even local maxima.



To generalize the single-peakedness idea to $\mathbb{R}^n$, let us rule out that any quasiconcave function on a convex subset of its domain has more than one *strict* local maximum. Assume to the contrary that both $\mathbf{x}$ and $\mathbf{x}'$ are strict local maxima. Then, by quasiconcavity, their convex combination, $\mathbf{x}^t$, is weakly better than at least one of $\mathbf{x}$ and $\mathbf{x}'$, say $\mathbf{x}$. However, when t is small and $\mathbf{x}^t$ is close to $\mathbf{x}$, the last conclusion contradicts that $\mathbf{x}$ is a strict local maximum. Hence, there can be at most one strict local maximum. Moreover, if such a strict local maximum exists, it must be the unique global maximum. Otherwise, the same contradiction reappears.

In the above discussion, we distinguished between the domain of the function, X , and the feasible set or constraint set, C , in the optimization problem. Recall that

X must be convex for technical reasons: Our definitions of convexity/concavity and quasiconvexity/quasiconcavity makes sense only if X is convex. However, C is usually shaped by economic circumstances, which begs the question: When is it reasonable to assume that C is a convex set?

Example: In many economic problems, the feasible set, C , of actions or choices is restricted by a number of inequalities. For instance, assume that X is convex and that the economic agent can choose only among $\mathbf{x} \in X$ that simultaneously satisfy

$$\begin{aligned} h(\mathbf{x}) &\leq a \\ g(\mathbf{x}) &\leq b, \end{aligned}$$

where $h(\mathbf{x})$ and $g(\mathbf{x})$ are functions with domain X and a and b are real numbers. Note that $\mathbf{x}$ must be in the inferior set of both $h(\mathbf{x})$ and $g(\mathbf{x})$. In other words, $\mathbf{x}$ must be in the intersection of the two inferior sets. Assume that $h(\mathbf{x})$ and $g(\mathbf{x})$ are quasiconvex. Then, the inferior sets are convex, and it follows that their intersection is a convex set as well. Thus, the feasible set is a convex set. Similarly, if the set of inequalities is written as

$$\begin{aligned} h(\mathbf{x}) &\geq a \\ g(\mathbf{x}) &\geq b, \end{aligned}$$

then the feasible set is convex if $h(\mathbf{x})$ and $g(\mathbf{x})$ are quasiconcave. Recall that linear functions are both quasiconvex and quasiconcave. Hence, any feasible set that is described only by linear inequalities is convex.

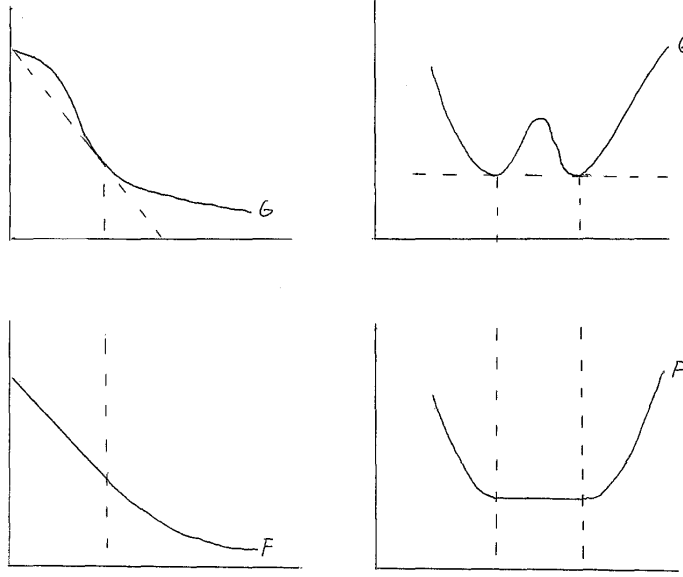
Exercise: Consider an objective function $f(\mathbf{x})$ that is strictly quasiconvex. Use a proof by contradiction to prove that any maximum (if one exists) is at an extreme point of the domain $X \subseteq \mathbb{R}^n$. The solution is at the end of the appendix.

4.2 Convexification and concavification (advanced)

There are applications in which the function under consideration is not convex or concave, but where it turns out to be worthwhile to examine the “closest” convex or concave function. We start with some function $g : X \rightarrow \mathbb{R}$, where X is convex. Then, for a non-convex function g , we can find *the highest convex function that is always weakly below g* . The figures on the next page provide some examples and illustrations.

To be more systematic, note that the problem with g is that its epigraph is not a convex set. However, we know how to construct the convex hull of such a set. The function f which has the latter as its epigraph must by definition be convex. Thus, with some abuse of terminology, we say that the convex hull of a non-convex function g is the function

$$f(\mathbf{x}) = \inf\{y | (\mathbf{x}, y) \in \text{co}(\text{epi}(g))\}.$$



Alternative terminology is that we have “convexified” the function g . Examining the figures, you see that the epigraph of f is precisely the convex hull of the epigraph of g .¹ Since $\text{co}(\text{epi}(g))$ is the set of all the convex combinations of points in $\text{epi}(g)$, f can also be obtained as the solution to a minimization problem, or

$$f(\mathbf{x}) = \inf \left\{ \sum_{i=1}^m t_i g(\mathbf{x}_i) \mid \sum_{i=1}^m t_i \mathbf{x}_i = \mathbf{x}, \text{ with } t_1, t_2, \dots, t_m \geq 0 \text{ and } \sum_{i=1}^m t_i = 1 \right\}.$$

Provided that g has a minimum, it can be shown that the minimum value of g and f coincide, or $\min_{\mathbf{x} \in X} g(\mathbf{x}) = \min_{\mathbf{x} \in X} f(\mathbf{x})$.² Moreover, $\arg \min_{\mathbf{x} \in X} g(\mathbf{x})$ is a subset of $\arg \min_{\mathbf{x} \in X} f(\mathbf{x})$. The second figure above illustrates these points.

If we are trying to “concavify” g instead, then we will look to find the smallest concave function that is always weakly above g . We can do so by constructing the

¹The reason for the use of $\inf$ rather than $\min$ in this definition (and the following) is slightly subtle. To understand it, consider the following non-convex function, with $X = \mathbb{R}_+$:

$$g(x) = \begin{cases} x & \text{if } x \in [0, 1) \\ \frac{1}{x} & \text{if } x \in [1, \infty) \end{cases}.$$

We want to construct the convex hull of the epigraph, so we take the convex combination of all points in the epigraph. Consider the following two points on the boundary of the epigraph, $(0, g(0)) = (0, 0)$ and $(x, g(x)) = (x, \frac{1}{x})$ for $x \geq 1$. When $t = \frac{x-1}{x}$, the convex combination is $(1, \frac{1}{x^2})$, which, again, must be in the convex hull. However, as $x \rightarrow \infty$, $\frac{1}{x^2} \rightarrow 0$. Hence, the infimum is zero, but the minimum does not exist (the convex hull is not a closed set since it does not contain all of its boundary). In this example, $f(x) = 0$ for all $x \in X$. Similar examples can be constructed when the domain is compact but the function is not continuous.

²Let $g^* = \min_{\mathbf{x} \in X} g(\mathbf{x})$. Since f is always weakly below g , $\min_{\mathbf{x} \in X} f(\mathbf{x}) \leq \min_{\mathbf{x} \in X} g(\mathbf{x}) = g^*$. The function $h(\mathbf{x}) = g^*$ is convex and weakly below g . However, since f is by definition the highest such function, $f(\mathbf{x}) \geq h(\mathbf{x})$ for all $\mathbf{x} \in X$. Thus, $\min_{\mathbf{x} \in X} f(\mathbf{x}) \geq \min_{\mathbf{x} \in X} h(\mathbf{x}) = g^*$. This proves that $\min_{\mathbf{x} \in X} f(\mathbf{x}) = \min_{\mathbf{x} \in X} g(\mathbf{x})$.

convex hull of the hypograph, or by solving a maximization problem that is analogous to the minimization problem mentioned above.

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Appendix: Some basic definitions

A.1 Open, closed, and compact sets

We consider sets in $\mathbb{R}^n$. As before, let $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ and $\mathbf{x}' = (x'_1, x'_2, \dots, x'_n) \in \mathbb{R}^n$ be two points in $\mathbb{R}^n$. We first wish to build a sense of distance between the two points. Although there are different ways to measure distance, we make use of the most intuitive and common measure. Thus, let

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2},$$

denote the “Euclidian norm.” It measures the distance between $\mathbf{x}$ and the origin. Next, let

$$\|\mathbf{x} - \mathbf{x}'\| = \sqrt{(x_1 - x'_1)^2 + (x_2 - x'_2)^2 + \dots + (x_n - x'_n)^2}$$

denote the Euclidian *distance* between the two points. Note that $\|\mathbf{x} - \mathbf{x}'\| > 0$ whenever $\mathbf{x} \neq \mathbf{x}'$.

Open ball: Given some $\mathbf{x} \in \mathbb{R}^n$ and $\varepsilon > 0$, the set $B_\varepsilon(\mathbf{x}) = \{\mathbf{x}' \in \mathbb{R}^n \mid \|\mathbf{x} - \mathbf{x}'\| < \varepsilon\}$ is an open ball around $\mathbf{x}$ (with radius ε).

First, a reminder of the (hopefully familiar) definitions of open and closed sets. We start with a set, $X \subseteq \mathbb{R}^n$, to which we compare some subset $S \subseteq X$. For this reason, the formal definitions mention properties of S *relative to* X . Usually, the last part is omitted, in which case it is understood that X is taken to be $\mathbb{R}^n$.

Open set: Fix a set $X \subseteq \mathbb{R}^n$. A set $S \subseteq X$ is said to be *open* relative to X if for every $\mathbf{x} \in S$ there exists some $\varepsilon > 0$ such that $B_\varepsilon(\mathbf{x}) \cap X \subseteq S$. That is, all the elements of the open ball around $\mathbf{x}$ that are also in X must belong to S .

Closed set: Fix a set $X \subseteq \mathbb{R}^n$. A set $S \subseteq X$ is said to be *closed* relative to X if its complements $X \setminus S$ is open relative to X .

A set need not be either open or closed. For example, $S = (0, 1]$ is not open (consider $x = 1$) nor closed (consider $x = 0$) relative to $X = \mathbb{R}$. However, $S = (0, 1]$ is open relative to $X = (-\infty, 1]$ (since at $x = 1$, $B_\varepsilon(1) \cap X = (1 - \varepsilon, 1] \subseteq S$ when $\varepsilon > 0$ is small). This last example illustrates the role of X . However, in the following we make the common assumption that $X = \mathbb{R}^n$ and no longer make any explicit reference to X . In this case, the set S is open if for every $\mathbf{x} \in S$ there exists some $\varepsilon > 0$ such that $B_\varepsilon(\mathbf{x}) \subseteq S$.

It is often important to distinguish between sets that are bounded and those that are not.

Bounded set: A set $S \subseteq \mathbb{R}^n$ is bounded if there exists some $r > 0$ such that $\|\mathbf{x}\| < r$ for all $\mathbf{x} \in S$.

Compact set: A set $S \subseteq \mathbb{R}^n$ is compact if it is closed and bounded.

A.2 The interior of a set and its closure and boundary

We move on to introduce ways to distinguish between different points inside or outside the set.

Limit point: A point $\mathbf{x} \in \mathbb{R}^n$ is a limit point of $S \subseteq \mathbb{R}^n$ if every open ball around $\mathbf{x}$ has elements distinct from $\mathbf{x}$ that are in S .

Note that the limit point does not itself have to be a member of S . The set S is closed if and only if it contains all of its limit points.

Interior points and the interior of a set: A point $\mathbf{x} \in S$ is an interior point if there exists $\varepsilon > 0$ such that $B_\varepsilon(\mathbf{x}) \subseteq S$. The interior of the set S , $\text{int}S$ is the set of interior points, or $\text{int}S = \{\mathbf{x} \in S \mid \text{there exists } \varepsilon > 0 \text{ such that } B_\varepsilon(\mathbf{x}) \subseteq S\}$. Note that $\text{int}S$ is an open set. It is the largest open subset contained in S .

Closure of a set: The closure of the set S , denoted $cl(S)$, is the union of the set S and its limit points. This is, as the name suggests, a closed set. Indeed, it is the smallest closed set containing S .

Boundary points and the boundary of a set: The boundary of set S is the (closed) set $cl(S) \setminus \text{int}S$, i.e. the set of points in the closure of S that are not interior. An element of this set is a boundary point.

EXAMPLE: Let $S = \{0\} \cup (1, 2]$. We note that 0 is not a limit point. However, every point in $[1, 2]$ is a limit point (including 1 even though $1 \notin S$). Thus, $cl(S) =$

$\{0\} \cup [1, 2]$. In comparison, $\text{int} S = (1, 2)$. The boundary of the set is $\{0\} \cup \{1\} \cup \{2\}$. Note that the closure and the boundary are closed set, as their complements are open sets.

A.3 Separating hyperplanes and support functions

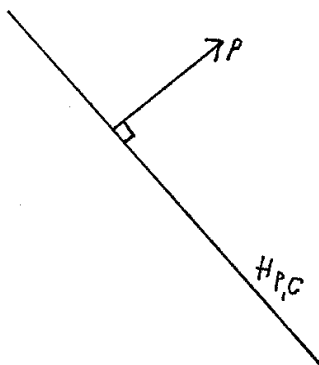
We begin by defining some special convex sets.

Hyperplanes: For any $c \in \mathbb{R}$ and $\mathbf{p} \in \mathbb{R}^n$ with $\mathbf{p} \neq \mathbf{0}$, the hyperplane generated by $\mathbf{p}$ and c is the set $H_{\mathbf{p},c} = \{\mathbf{x} \in \mathbb{R}^n | \mathbf{p} \cdot \mathbf{x} = c\}$.

The vector $\mathbf{p}$ is the *normal vector* of the hyperplane. Understanding why is helpful for geometric illustrations of the following concepts and results. Hence, take some focal point $\mathbf{x}^0 \in H_{\mathbf{p},c}$ and compare it to other $\mathbf{x} \in H_{\mathbf{p},c}$. By definition, $\mathbf{p} \cdot \mathbf{x}^0 = \mathbf{p} \cdot \mathbf{x}$, so we can also define $H_{\mathbf{p},c}$ as

$$H_{\mathbf{p},c} = \{\mathbf{x} \in \mathbb{R}^n | \mathbf{p} \cdot (\mathbf{x} - \mathbf{x}^0) = 0\}.$$

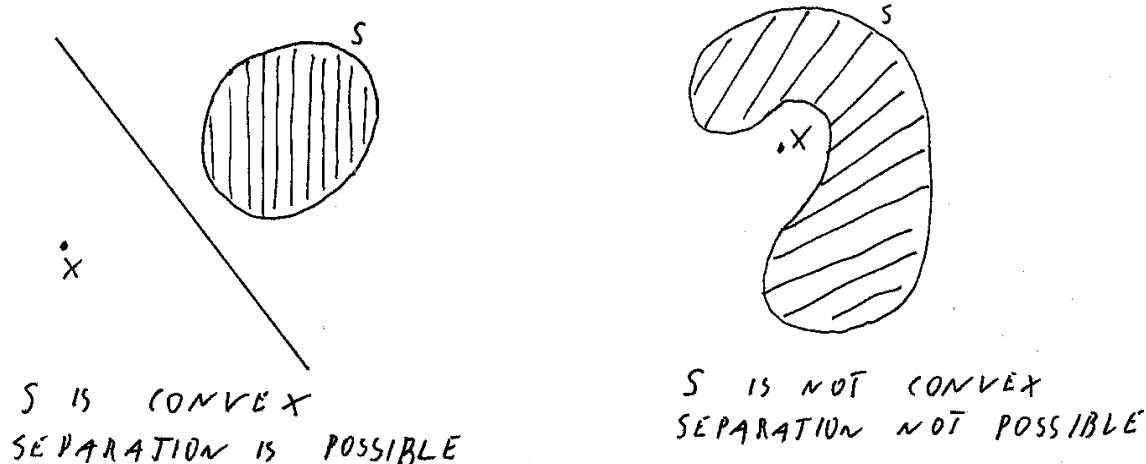
Recall that the dot product of two vectors is zero only if the two vectors are orthogonal. That is, $\mathbf{p}$ is perpendicular or normal to the hyperplane. This explains why the hyperplane is often depicted as follows:



Half-spaces: For any hyperplane, the set $\{\mathbf{x} \in \mathbb{R}^n | \mathbf{p} \cdot \mathbf{x} \geq c\}$ is the (closed) half-space above the hyperplane and the set $\{\mathbf{x} \in \mathbb{R}^n | \mathbf{p} \cdot \mathbf{x} \leq c\}$ is the (closed) half-space below the hyperplane.

Clearly, hyperplanes and half-spaces are very special cases of convex sets. We want to use their special properties to explore other sets, especially convex sets. We begin with separating hyperplane theorems. The central question is: Given some set S , and a point $\mathbf{x}$ outside S , when is it the case that there exists a hyperplane that splits $\mathbb{R}^n$ in such a way that $\mathbf{x}$ is in one half-space and S in the other? Note that this is an *existence* question. Thus, separating hyperplane theorems are often invoked in different existence theorems (we will use it for this purpose in General Equilibrium).

As the following figures illustrate, it is easy to see that we should not expect separation to be always possible if S is non-convex. Thus, we focus on convex S , in which case it is intuitive that separation is possible. However, the degree of separation depends on whether S is closed or not. The following break-down of separation theorems is taken from Vohra (2005).



Strict separating hyperplane theorem: Let $S \subseteq \mathbb{R}^n$ be a closed convex set and $\mathbf{x} \notin S$. Then, there exist $c \in \mathbb{R}$ and $\mathbf{p} \in \mathbb{R}^n$ with $\mathbf{p} \neq \mathbf{0}$ such that $\mathbf{p} \cdot \mathbf{x} > c$ and $\mathbf{p} \cdot \mathbf{y} < c$ for all $\mathbf{y} \in S$.

Weak separating hyperplane theorem: Let $S \subseteq \mathbb{R}^n$ be a convex set and $\mathbf{x} \notin S$. Then, there exist $c \in \mathbb{R}$ and $\mathbf{p} \in \mathbb{R}^n$ with $\mathbf{p} \neq \mathbf{0}$ such that $\mathbf{p} \cdot \mathbf{x} \geq c$ and $\mathbf{p} \cdot \mathbf{y} \leq c$ for all $\mathbf{y} \in S$.

Rather than separating a point from a set, we sometimes wish to separate two arbitrary convex sets. This gives us the following counterparts to the above theorems.

Strict separation between convex sets: Let S and T be two non-empty, disjoint, closed, convex sets in $\mathbb{R}^n$, with S bounded. Then, there exist $c \in \mathbb{R}$ and $\mathbf{p} \in \mathbb{R}^n$ with $\mathbf{p} \neq \mathbf{0}$ such that $\mathbf{p} \cdot \mathbf{x} > c > \mathbf{p} \cdot \mathbf{y}$ for all $\mathbf{x} \in S$ and $\mathbf{y} \in T$.

Weak separation between convex sets: Let S and T be two non-empty, disjoint, convex sets in $\mathbb{R}^n$. Then, there exist $c \in \mathbb{R}$ and $\mathbf{p} \in \mathbb{R}^n$ with $\mathbf{p} \neq \mathbf{0}$ such that $\mathbf{p} \cdot \mathbf{x} \geq c \geq \mathbf{p} \cdot \mathbf{y}$ for all $\mathbf{x} \in S$ and $\mathbf{y} \in T$.

Returning to the original separation theorems, any $\mathbf{x} \notin S$ is in some open half-space with S in the complementary closed half-space. Hence, if we take the intersection of all the closed half-spaces that contain S , then we must have removed all points not in S . In other words, a closed convex set is the intersection of the closed half-spaces that contain it. More generally, if S is not convex, then the closure of its convex hull is the intersection of all the closed half-spaces that contain S .

We are now prepared to discuss the support function and the duality theorem that is presented in MWG Section 3.F. We begin with the observation that there are of course many half-spaces that contain S . Indeed, even if we fix $\mathbf{p}$, then there are still many c for which S is in the relevant half-space. More concretely, if $\mathbf{p} \cdot \mathbf{x} \geq c$ for all $\mathbf{x} \in S$ then it is also true that $\mathbf{p} \cdot \mathbf{x} \geq c'$ for all $\mathbf{x} \in S$ when $c' \leq c$. Thus, there is a lot of redundancy, and it seems to make sense to ask what the smallest half-spaces that contain S are. To this end, fix $\mathbf{p}$ and solve for the number

$$\mu_S(\mathbf{p}) = \inf\{\mathbf{p} \cdot \mathbf{x} | \mathbf{x} \in S\},$$

which is the smallest possible value that $\mathbf{p} \cdot \mathbf{x}$ can take in S . If we let c take this value, or $c = \mu_S(\mathbf{p})$, then it is certainly true that $\mathbf{p} \cdot \mathbf{x} \geq c \geq c'$ for all $\mathbf{x} \in S$ and all $c' \leq c$. In other words, we have identified the smallest half-space with normal vector $\mathbf{p}$ that contains S . We can do this for every value of $\mathbf{p}$, which is why $\mu_S(\mathbf{p})$ is written as a function of $\mathbf{p}$. This leads us to the following definition.³

Support functions⁴: Let $S \subseteq \mathbb{R}^n$ be a non-empty and closed set. The support function of S is defined, for all $\mathbf{p} \in \mathbb{R}^n$, as

$$\mu_S(\mathbf{p}) = \inf\{\mathbf{p} \cdot \mathbf{x} | \mathbf{x} \in S\}.$$

The punch-line is that if we know the support function $\mu_S(\mathbf{p})$ and that S is closed and convex, then we have enough information to completely recover S . In this sense, the support function gives a “dual” or equivalent way of characterizing S . In fact, we already explained the intuition: $\mu_S(\mathbf{p})$ gives us the smallest half-spaces that contain S , and we therefore know that anything not in the convex set S has been excluded. To be specific, S coincides with the set

$$\{\mathbf{x} \in \mathbb{R}^n | \mathbf{p} \cdot \mathbf{x} \geq \mu_S(\mathbf{p}) \text{ for all } \mathbf{p} \in \mathbb{R}^n\}.$$

If S is not convex, then the above set is the closure of the convex hull of S .

The support function has a few interesting properties. First, it is homogeneous of degree one in $\mathbf{p}$. Second, it is concave because (using $\mathbf{p}^t$ to denote the convex combination of $\mathbf{p}$ and $\mathbf{p}'$)

$$\begin{aligned} \mu_S(\mathbf{p}^t) &= \inf\{\mathbf{p}^t \cdot \mathbf{x} | \mathbf{x} \in S\} \\ &= \inf\{(t\mathbf{p} + (1-t)\mathbf{p}') \cdot \mathbf{x} | \mathbf{x} \in S\} \\ &\geq \inf\{t\mathbf{p} \cdot \mathbf{x} | \mathbf{x} \in S\} + \inf\{(1-t)\mathbf{p}' \cdot \mathbf{x} | \mathbf{x} \in S\} \\ &= t\mu_S(\mathbf{p}) + (1-t)\mu_S(\mathbf{p}'). \end{aligned}$$

³This definition of the support function follows MWG. In other sources, half-spaces of the form $\mathbf{p} \cdot \mathbf{x} \leq c$ are considered instead, and $\inf$ is therefore replaced with $\sup$ in the definition.

⁴The name reflects the fact that the support function is related to supporting hyperplanes and supporting half-spaces, the latter of which is a half-space that contains S and which has a point of S in its boundary

Finally, note that there may be multiple solutions to $\inf_{\mathbf{x} \in S} \mathbf{p} \cdot \mathbf{x}$, i.e. multiple values of $\mathbf{x}$ for which $\mathbf{p} \cdot \mathbf{x}$ is minimized. Whether this is the case or not evidently depends on the general shape of S . If S is *strictly* convex, then not only is the solution unique for every $\mathbf{p} \neq \mathbf{0}$, the support function is also differentiable in $\mathbf{p}$.⁵ Indeed, it then holds that the unique solution to $\inf_{\mathbf{x} \in S} \mathbf{p} \cdot \mathbf{x}$ coincides with $\nabla \mu_S(\mathbf{p})$. The converse also holds. That is, if $\mu_S(\mathbf{p})$ is differentiable, then the solution is unique. MWG refers to these results as the Duality Theorem. Note that the problem $\inf_{\mathbf{x} \in S} \mathbf{p} \cdot \mathbf{x}$ treats $\mathbf{x}$ as the variables, whereas when we find $\nabla \mu_S(\mathbf{p})$ it is as if we treat $\mathbf{p}$ as the variables.

Shephard's lemma can be seen as an implication of the duality theorem. Consider a locally non-satiated and strictly convex preference relation, represented by utility function $u(\mathbf{x})$. Consider the superior set $S = \{\mathbf{x} \in \mathbb{R}_+^n | u(\mathbf{x}) \geq u\}$. Then, the expenditure function is precisely the support function to S , or $\mu_S(\mathbf{p}) = e(\mathbf{p}, u)$! Since preferences are locally non-satiated and strictly convex, the expenditure minimization problem has a unique solution, $\mathbf{h}(\mathbf{p}, u)$ for any $\mathbf{p} \gg \mathbf{0}$. By the duality theorem, it thus holds that the gradient of the support function coincides with the solution, or $\frac{\partial e(\mathbf{p}, u)}{\partial p_i} = h_i(\mathbf{p}, u)$, which is precisely Shephard's lemma. Similarly, since $e(\mathbf{p}, u)$ is the support function, it is homogeneous of degree one in $\mathbf{p}$ and concave in $\mathbf{p}$.

A.4 Solution to the exercise in Section 4.1

Exercise: Consider an objective function $f(\mathbf{x})$ that is strictly quasiconvex. Use a proof by contradiction to prove that any maximum (if one exists) is at an extreme point of the domain $X \subseteq \mathbb{R}^n$.

Solution: By contradiction, assume that there is a solution, $\mathbf{x} \in X$, that is *not* an extreme point of X . However, this implies that $\mathbf{x}$ is a convex combination of two distinct points in X , or $\mathbf{x} = t\mathbf{x}' + (1 - t)\mathbf{x}''$ for some $\mathbf{x}', \mathbf{x}'' \in X$ with $\mathbf{x}' \neq \mathbf{x}''$ and some $t \in (0, 1)$. Since the function is strictly quasiconvex

$$f(\mathbf{x}) < \max\{f(\mathbf{x}'), f(\mathbf{x}'')\},$$

but this contradicts that $\mathbf{x}$ is a maximum. Hence, any maximum must be an extreme point.

⁵Note that we are not imposing differentiability of the function $\mu_S(\mathbf{p})$ as an assumption. Instead, it emerges as a conclusion, which is derived from an underlying property of the set S .